INCH-POUND

MIL-C-44417(GL)
21 February 1991

MILITARY SPECIFICATION

CARRIER, BODY ARMOR, SMALL ARMS PROTECTIVE,

AIRCREW, SURVIVAL

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a carrier for a ceramic body armor insert to be worn by aircrew personnel. The carrier is part of an aircrew survival system. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The carrier shall be of the following types and sizes as specified (see 6.2).

Type I - Carrier
Type II - Carrier with ceramic body armor insert

Schedule of sizes

Short	Medium	Long	X-Long

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8470

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-T-295	-	Thread, Nylon
DDD-L-20	-	Label, For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-W-4088	-	Webbing, Textile, Woven Nylon
MIL-T-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-W-5664	-	Webbing, Textile, Elastic
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43128	-	Cloth, Plain Weave, Nylon: Water Repellent, OG-106
MIL-W-43668	-	Webbing, Textile, Textured or Multifilament
MIL-C-43842	-	Cloth, Oxford, Aramid
MIL-C-44050	-	Cloth, Ballistic, Aramid
MIL-B-44421	-	Body Armor, Small Arms Protective, Aircrew, Survival

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement
FED-STD-751	-	Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-970	-	Standards and Specifications, Order of Preference for the Selection of

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-6-100 - Buckle, Side Release, 2 inch

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commerical Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

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3.3 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not specified shall be selected by the contractor in accordance with MIL-STD-970. Materials and components specified in accordance with MIL-STD-970 shall not degrade the operational suitability or effectiveness of the carrier. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Cloth, oxford, aramid. The oxford aramid cloth used for the body of the carrier shall conform to MIL-C-43842. The color shall be Olive Green 106.

3.3.2 Cloth, ballistic, aramid. The ballistic aramid cloth used for the insert shall conform to type II, class 2 of MIL-C-44050.

3.3.3 Thread, nylon. The thread used throughout the carrier shall be types I, II or III, class A, letter size E or FF conforming to V-T-295. The color shall be Camouflage Green 483 approximating color chip 34094 of FED-STD-595.

3.3.4 Fastener tape, hook and loop, synthetic. The fastener tape shall be type I, class 1 conforming to MIL-F-21840. The widths shall be as specified in table I. The color shall be Olive Green 106.

3.3.5 Webbing, textile, woven nylon. The nylon webbing used for the harness and pull strap shall be type VIII, class 1 conforming to MIL-W-4088. The color shall be Olive Drab 7.

3.3.6 Webbing, textile, elastic. The elastic webbing used for the side waist adjustment straps shall be type II, class 1, 1-1/2 inches wide conforming to MIL-W-5664. The color shall be Olive Green 106.

3.3.7 Webbing, textile, textured or multifilament nylon. The webbing used on the harness support loop shall be type III conforming to MIL-W-43668. The color shall be Olive Drab 7.

3.3.8 Buckle, side release, 2 inch. The buckle used for securing the harness shall conform to Drawing 2-6-100.

3.3.9 Tape and webbing, nylon. The tape used for the harness support loop and binding the ballistic cloth insert shall be type IV, class 2, 1- inch wide conforming to MIL-T-5038. The color shall be Olive Drab 7.

3.3.10 Labels. Each carrier shall have a combination size, identification and instruction label positioned on the inside of the back. The label shall conform to type VI, class 14 of DDD-L-20 with the following exceptions and detailed requirements.

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a. The size characters for the instruction label, except capital letters, shall be 1/16 inch minimum. All capital letters shall be 3/32 inch minimum.

b. The label material shall be nylon conforming to MIL-C-43128, except it shall not be water-repellent treated. The label material shall be dyed Medium Green 614 matching the standard sample (see 6.4).

c. The required permanent coating shall not apply.

d. The label shall show fastness to laundering in accordance with class 14 of DDD-L-20.

e. The instruction label shall contain the following information:

READ THESE INSTRUCTIONS CAREFULLY

WASHING INSTRUCTIONS FOR CARRIER, BODY ARMOR, SMALL ARMS PROTECTIVE,

AIRCREW, SURVIVAL

1. Hand/Machine Washing. Hand wash in warm water (120°F) and mild laundry detergent or machine wash at permanent press cycle.

2. Rinsing. Rinse thoroughly in clear warm water after laundering. All detergents must be removed. After rinsing, hang the carrier to drip dry or tumble dry with the temperature dial set at permanent press cycle.

3. Fixed laundering procedure. Wash the carrier according to laundry wash formula "K" of FM-10-17. Lightly extract, then tumble dry at 180°F (82°C) maximum.

4. Field laundry procedure. Wash the carrier according to laundry wash formula "1" of TM 10-280. Lightly extract, then tumble dry at 180°F (82°C) maximum.

5. DO NOT STARCH OR USE CHLORINE BLEACH UNDER ANY CONDITIONS.

6. DO NOT DRY CLEAN. DO NOT REMOVE THIS LABEL.

7. WARNING: Launder with the hook and loop fastener tapes engaged. Do not press hook and loop fasteners.

3.3.11 Identification marking. The words "DO NOT DROP" and "WARNING: ENSURE HOOK AND LOOP ARE PROPERLY FASTENED" shall be printed in characters 1 inch high on the outside front of the carrier, so as to be visible when worn. The markings shall conform to type IV, class 9 of DDD-L-20. Fastness to class 9 markings shall be as specified for class 5.

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3.3.12 Ceramic body armor insert. The ceramic body armor insert for inclusion in the type II carrier (see 5.1.1.2) shall be of the same size as the carrier and shall conform to MIL-B-44421.

3.4 Design. The type I carrier consists of an aramid oxford cloth outer shell with aramid ballistic cloth insert. The carrier is open at one end to allow the insertion of the ceramic body armor insert. This opening is provided with a flap secured by a hook and loop tape fastener, which also acts as a quick release mechanism. The carrier is supported by a cross harness which consists of nylon webbing and quick release buckles. The type II carrier is identical to the type I carrier except that it includes the ceramic body armor insert.

3.5 Patterns. Standard patterns, to be used for cutting the contractor's working patterns, will be furnished by the Government. The standard patterns shall not be altered in any way and are to be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns. The patterns provide a 3/8 inch seam allowance. Pattern pieces requiring overedging may be cut oversized and trimmed to the finished size simultaneously with overedging as long as the finished component conforms to the required configuration.

3.5.1 List of pattern parts. The components of the carrier shall be cut from the material as specified in accordance with the pattern parts indicated below.

List of pattern parts

Material	Nomenclature	Parts
Ballistic cloth, aramid	Insert	7
Oxford cloth, aramid	Front	1
	Back	1
	Flap	1
	Side closure	1

3.6 Construction. The construction shall conform in all respects to the requirements specified in table I and herein.

3.6.1 Stitches, seams and stitching. All stitches, seams, stitching types and stitches per inch specified in table I shall conform to FED-STD-751. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers or open seams will result. When two or more methods of seam or stitch types are given for the same operation, any one may be used. The height of overedge stitching shall be 3/16 to 1/4 inch. Unless otherwise specified, edge stitching shall be 1/4 to 5/16 inches from edge.

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3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched no less than 1/4 inch except where ends are turned under in a hem or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.2 Bartacks. Bartacks shall measure 1/2 ± 1/16 inch and shall be free from thread breaks and loose stitching.

3.6.3 Thread ends. All thread ends shall be trimmed to 1/4 inch length. As an alternate, the thread ends may be heat fused.

3.6.4 Fusing of ends of nylon tape and webbing. All ends of nylon tape and webbing shall be fused. The apparatus used to fuse the tape ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the tape yarns fused together.

3.7 Fit. The finished carrier shall accommodate the ceramic body armor insert of applicable size for which it was designed, without strain on seams or excessive fullness.* The ceramic body armor insert required for fit examination shall be obtained from the contracting activity (see 6.4). DELETE

3.8 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the carrier.

3.9 Manufacturing operations requirements. The carrier shall be manufactured in accordance with operations specified in table I. The contractor is not required to follow the exact sequence of operations. Any additional basting or holding stitches used to facilitate manufacture is permissible provided the thread is removed or does not show on the finished carrier.

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3.10 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table I, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government furnished shaper pattern is forwarded, the component shall conform to that pattern.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. The oxford cloth for the carrier shall be cut in strict accordance with patterns furnished which show size, shape, directional lines and placement of components. b. The ballistic cloth for the insert shall be cut in strict accordance with patterns furnished which show size, shape and directional lines. c. No components will be drilled for positioning. d. The webbing for the harness (two for each carrier) shall be cut in the following lengths. The ends of the webbing shall be fused (see 3.6.4). The tolerance shall be ± 1/4 inch. Short - 130 inches Long - 134 inches Medium - 132 inches X-Long - 136 inches e. Cut the webbing for the pull strap (one for each carrier) 19 ± 1/8 inches long for all sizes. The ends shall be fused. f. Cut the webbing for the side waist adjustment straps (two for each carrier) 5 ± 1/8 inches long for all sizes. The ends shall be fused. g. Cut the webbing for the harness support loop 9 ± 1/8 inches long (one for each carrier) for all sizes. The ends shall be fused.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) h. Cut the 2 inch wide hook portion of the fastener tape 8 ± 1/16 inches in length (one for each carrier). i. Cut the 1-1/2 inch wide loop portion of the fastener tape 8 ± 1/16 inches in length (two for each carrier).						
2.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.						
3.	Marking. Mark, ticket or bundle all component parts to ensure a uniform shade throughout the carrier. Any method of marking may be used except: a. Metal fastening devices b. Sew-on tickets c. Adhesive type tickets which leave traces of adhesive on the material after removal of the tickets.						
4.	Preparation and assembly of the ballistic cloth insert. a. The seven plies of the ballistic cloth for the insert shall be evenly superimposed with all the	301	BSc-1	8-10	FF	FF	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Preparation and assembly of the ballistic cloth insert. (cont'd)						
	edges even. Simultaneously, join the plies together and bind the edges with the binding tape specified in 3.3.9. Turn binding tape over onto opposite side and stitch around entire periphery with a single row of stitching located 3/8 inch minimum from the edge of the ballistic cloth of the insert. The edge of the binding tape shall finish 1/2 inch minimum from the edge of the ballistic cloth of the insert. Ends of binding tape shall be overlapped 1 inch minimum and end of binding tape on outside shall be turned under 1/2 inch minimum.						
	or						
	b. Bind periphery of the ballistic cloth insert in two operations. Align the edge of binding tape even with edge of ballistic cloth insert and stitch all around periphery with stitching located 3/8 inch minimum from edge of insert. Turn binding over onto opposite side and stitch around periphery of insert with stitching located 3/8 inch minimum from the edge of the ballistic insert. The edge of the binding tape shall finish 1/2 inch minimum from the edge of the insert. Ends of binding tape shall be overlapped 1 inch minimum and the end of the binding tape on the outside shall be turned under 1/2 inch minimum.	301	BSg-2	8-10	FF	FF	
5.	Assemble side waist adjustment straps and harness.						
	a. Thread one end of the webbing for the side waist adjustment strap through the female portion of the	301		10-12	E	E	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Assemble side waist adjustment straps and harness. (cont'd)						
	buckle. With ends even, stitch 3/4 ± 1/16 inch away from buckle. Repeat the procedure for the other adjustment strap.						
	b. On one of the pieces of webbing for the harness, take the 9 inch long piece of webbing for the harness support loop and stitch it to the webbing 9 ± 1/8 inches from the end with a 1 inch boxstitch at each end.	301		10-12	E	E	
	c. Thread one end of the webbing for the harness through the male portion of the buckle. Turn under end 3/4 ± 1/16 inch, then turn onto itself another four times and boxstitch with a 1-1/2 by 1 inch box-stitch.	301		10-12	E	E	
6.	Assembling of the carrier.						
	a. Overedge the raw edges of the oxford cloth carrier pieces (front, back and side closure), except for the bottom edge of the carrier.	502, 503, 504 or 505	EFd-1	6-10	FF	FF	
	b. Stitch the side closure to the back with a 3/8 inch seam, catching the side waist adjustment straps and harness in the seam. Locate the side waist adjustment straps and harness as indicated on patterns, allowing 1-1/4 inches of webbing to be on the inside.	301	SSa-1	8-10	E	E	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Assembling of the carrier. (cont'd)						
	Turn the carrier right side out and place a 1-1/2 by 1 inch boxstitch through the side closure and webbing for the side waist adjustment strap and harness (in four places).						
	c. Stitch the other end of the side closure to the front with a 3/8 inch seam. Turn the carrier right side out.	301	SSa-2	8-10	E	E	
7.	Hem carrier.						
	a. Turn up bottom edge of carrier 1/4 inch and then 1/4 inch again and hem. The hem shall finish 1/4 ± 1/16 inch.	301	EFb-1	8-10	E	E	
	b. Topstitch the seam of the side closure and front. The stitch margin shall be 1/8 ± 1/16 inch.	301		8-10	E	E	
8.	Attach loop fastener tape to back of carrier.						
	a. On outside of back attach two pieces of 1-1/2 inch loop fastener tape according to marks on pattern. Each piece of loop tape shall be 8 ± 1/16 inches long.						
	b. Each piece of loop tape shall be stitched on all four sides with a 1/8 ± 1/16 inch seam.	301	EFb-1	8-10	E	E	
9.	Make and attach flap.						
	Finished appearance. The flap shall be uniformly stitched and shall finish smooth and flat without twists, gathers, puckers, pleats or raw edges.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Make and attach flap. (cont'd)						
	a. Fold flap in half, with right sides together and stitch 1/4 ± 1/16 inch from the edges, leaving the top open.	301	SSe-2(a)	10-14	E	E	
	b. Turn flap right side out and work out edges. Top stitch side and bottom edges 1/4 ± 1/16 inch from edges.	301	SSe-2(b)	10-14	E	E	
	c. Attach 2 inch wide hook fastener tape to flap with pull strap placed under hook tape as indicated on pattern. The hook tape shall be 8 ± 1/16 inches long.						
	d. Stitch hook tape on all four sides with a 1/8 ± 1/16 inch seam.	301	SSe-1	8-10	E	E	
	e. With flap placed on carrier front, as indicated on pattern, and raw edge of flap lined up with the finished edge of carrier stitch 3/16 to 1/4 inch from edge. Flip over on itself and topstitch to carrier 3/8 inch (only the length of the flap).	301	SSe-1	8-10	E	E	
	f. Turn pull strap to outside of flap and topstitch 1/4 ± 1/16 inch from edge. On the other end of the pull strap fold over 5-1/2 ± 1/8 inch and stitch at fused end of strap 1/4 ± 1/16 inch away from end. Fold top end over again 1 ± 1/8 inch and stitch 1/8 ± 1/16 inch from first fold.	301	SSe-1	8-10	E	E	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Attach label. a. Combination size, identification and instruction label. The label shall be placed on the inside of the back, 2 ± 1/4 inches from the bottom edge and centered from the sides. b. The label shall be sewn to the carrier on all four sides with stitching not more than 3/16 inch from edge of label. Stitching shall not be through the writing on the label.	301	LSbj-1	10-12	E	E	
11.	Insert ballistic cloth insert into carrier and secure. Evenly insert the ballistic cloth insert into the carrier. Make four 1/2 inch bartacks through the insert and front of the carrier, as indicated on patterns.	Bar-tack		28 per bartack	FF	FF	
12.	Clean carrier. Remove all spots, stains and size markings. All thread ends and any ballistic cloth yarns protruding from the binding shall be trimmed to a length of not more than 1/4 inch.						

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3.11 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Utmost care shall be taken during fabrication to ensure quality workmanship and safety of the service person using the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.3), it shall be examined as specified in 4.4.3 and 4.4.4.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of carriers. The sample unit shall be one carrier. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 6.5 for major and minor A combined defects, and 15.0 for total (major, minor A and minor B combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
Cutting	Any component part not cut in accordance with directional lines indicated on patterns, or not in accordance with specified requirements	101		
Fabric	Any hole, cut, tear, smash, or broken or missing yarn	102		
	Any abrasion mark			301
Fastener tape, hook and loop	Any cut, hole, or tear	103		
	Missing hooks	104		
	Hooks flattened	105		
	Hook and loop out of alignment or not able to be fastened properly		201	
	Set on crookedly			302
	Width of hook and loop not as specified			303
Webbing, nylon	Any hole, cut, tear or frayed edge	106		
	Ends not fused as specified	107		
	Not specified width		202	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Webbing, elastic	Cut, chafed, abraided, or otherwise damaged affecting serviceability	108		
Components and assembly	Any component or required operation omitted (unless otherwise specified herein)	109		
	Any component not as specified	110		
	Any component part twisted, distorted, pleated, misshaped, tight or full (unless otherwise specified)		203	
	Any operation not performed as specified (unless otherwise specified)			304
	Needle chews	111		
Seams and stitching	Accuarcy of seaming:			
	- Seam irregular, twisted, pleated or wavy			305
	NOTE: Pleats shall be scored when they exceed 1/4 inch in length and 1/16 inch in depth of fold.			
	- Seam puckered (score only when on major portion of seam)			306
	- Any part of carrier caught in an unrelated operation or stitching		204	
	- End of stitching when not caught in other seams or stitching backtacked less than 1/4 inch			307
	- Thread breaks (all stitch types) stitched less than 1/2 inch beyond each end of break			308
	- Ends of a continuous line of stitching not overlapped or overlapped less than 1/2 inch			309
	- Ends of stitching on label overlapped with less than three stitches			310

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Open seams:			
	- up to 1/2 inch inclusive			311
	- more than 1/2 inch		205	
	NOTE: A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.			
	Raw edges:			
	- On outside:			
	- up to 1/2 inch inclusive			312
	- more than 1/2 inch		206	
	- On inside, more than 1 inch			313
	NOTE: Raw edges not securely caught in stitching shall be classified as open seams.			
	Gage of stitching and seam allowance:			
	- Irregular, i.e., unevenly gaged, or corresponding stitchings not uniformly gaged (to be scored only when condition exists on more than 1/2 the length of the seam)			314
	- Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			315
	- Seam allowance not as specified or varies more than 1/8 inch			316
	Run-offs on edge stitching:			
	- 1/4 to 1/2 inch inclusive			317
	- more than 1/2 inch		207	
	Seam or stitch types:			
	- Not specified seam or stitch type		208	
	- Any line of stitching omitted			
	-affecting serviceability seriously	112		
	-affecting serviceability but not seriously		209	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Stitch tension:			
	- Loose tension, resulting in a loose seam:			
	- up to 1/2 inch inclusive			318
	- more than 1/2 inch		210	
	- Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	113		
	Stitches per inch (all seams):			
	- less than the minimum		211	
	- more than the maximum			319
	NOTE: Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the fabric in order to sew over heavy places, or in turning corners shall be classified as follows: (a) within the minor B defect classification - no defect (b) within the minor A defect classification - Minor B defect			
	Height of boxstitching less than specified by more than 1/8 inch	114		
Cleanness	Noticeable grease or oil stains			320
	Thread ends not trimmed throughout			321
Flap	Flap not formed as specified		212	
	Flap improperly set or distorted, failing to effect a full and smooth closure		213	
	Flap out of alignment with fastener tape by more than 1/2 inch			322
Bartacks	One or more bartacks omitted		214	
	Any bartack not as specified or not in specified location			323
	Loose, incomplete or broken stitching		215	
	Not specified size		216	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Reinforcements	Omitted or mislocated		217	
Marking	Omitted, incorrect, illegible, misplaced, not applied as specified or size of characters not as specified	115		
Hardware: General	Broken or malformed; corroded areas; burrs or sharp edges	116		
	Size or type not as specified	117		
	Not assembled as specified, improperly set or reversed, failing to serve intended purpose	118		
	Finish omitted or not as specified		218	
Label	Omitted, illegible, misplaced		219	
	Not type and class specified	119		
	Not positioned as specified			323
	Not stitched on all four sides			324
	Label stitching through the printing		220	
Hems	Less than specified width on hems		221	
Adjustment strap and buckle	Improperly threaded, i.e., causing buckle to finish on reverse side or to function improperly		222	
	Turned under edge of adjustment strap not as specified			325
	Buckle defective, affecting function		223	

4.4.4 End item examination for fit. The end items shall be tested for the defects listed in table III. The lot size shall be expressed in units of carriers. The sample unit shall be one carrier. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall 4.0.

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TABLE III. Fit defects

Examine	Defect	Classification Major	Minor
NOTE:	Fit examination shall be acomplished with the ceramic body armor insert for which the carrier is designed. The inspector shall ease the ceramic armor insert into the carrier without effort and close the opening securely with the touch and hold fastener tape.		
Fit of ceramic body armor insert in carrier (see 6.4)	Carrier too small, i.e., ceramic body armor insert fails to fit into carrier	101	
	Fastener tape fails to effect a smooth and secure closure		201
	Carrier too large, showing excessive fullness or wrinkles		202

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling
	Bulged or distorted container
Contents	Number per container is more or less than specified

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Carrier, when procured separately (type I). Each carrier shall be folded to measure approximately 17 inches by 14 inches. Each carrier shall be unit packed in a plastic bag conforming to A-A-50083.

5.1.1.2 Carrier, when procured with ceramic body armor insert (type II). The ceramic body armor insert conforming to MIL-B-44421 shall be placed in a carrier of the correct size and the flap secured.

5.1.2 Commercial preservation. Each carrier or carrier and body armor assembly shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commerical, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Carrier, when procured separately (type I). Twenty carriers of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Inside dimensions of each shipping container shall approximate 17-1/4 inches in length, 14 inches in width and 12 inches in depth. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping for type II, class F loads.

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5.2.1.2 Carrier, when procured with ceramic body armor insert (type II). Each carrier with ceramic body armor insert of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Inside dimensions of each shipping container for sizes small and medium shall approximate 14-1/2 inches in length, 12-1/2 inches in width and 8 inches in depth. Inside dimensions of the shipping container for sizes long and X-long shall approximate 15-1/4 inches in length, 14-1/2 inches width and 8-1/2 inches in depth. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping for type II, class F loads.

5.2.2 Level B packing.

5.2.2.1 Carrier, when procured separately (type I). Twenty carriers of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of each shipping container shall approximate 17-1/4 inches in length, 14 inches in width and 12 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.2 Carrier, when procured with ceramic body armor insert (type II). Each carrier with ceramic body armor insert of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of each shipping container for sizes short and medium shall approximate 14-1/2 inches in length, 12-1/2 inches in width and 8 inches in depth. Inside dimensions of each shipping container for sizes long and X-long shall approximate 15-1/4 inches in length, 14-1/2 inches in width and 8-1/2 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.3 Weather-resistant shipping container. When specified (see 6.2), shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. Carriers (type I) or carriers with ceramic body armor insert (type II), preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), carriers (type I) or carriers with ceramic body armor insert (type II), packed as specified in 5.2.2 or 5.2.3 shall be palletized on a 4-way entry pallet in accordance with

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load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 95 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The carrier with a ceramic body armor insert is intended to be worn over the chest area to provide protection from small arms fire for aircrew members exposed to survival and recovery procedures.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Type, class and size required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.2, 4.3 and 6.3).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Type and class of unit load required (see 5.2.1.1 and 5.2.1.2).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).
i. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

CHANGE 6.4 Sample. For access to samples and to ceramic body armor inserts for the fit examination of type I carriers, address the contracting activity issuing the invitation for bids or request for proposal.

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6.5 Subject term (key word) listing.

Ballistic protection
Ceramic body armor insert
Life Support Clothing and Equipment
LSC&E
SARVIP system

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8470-0140)

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